

Al/Cu Enhancement in Marine Anti-Biofouling and Anti-Biocorrosion Performance of High-Entropy Alloys

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Antibacterial alloys are designed to mitigate biofouling, which is a strategy for combining the defense and function properties of alloys in complex biological environments. However, the current approaches are unable to overcome the issue of microbiologically influenced corrosion (MIC), because antibacterial performances mostly benefit from the excessive dissolution of antibacterial elements in alloys. Herein, a series of $\text{Al}_x\text{CoCrCuFeNi}$ high-entropy alloys (molar fractions $x = 0, 0.1, 0.3, 0.5$) are designed. Benefiting from the rigorous management of Al and Cu components, $\text{Al}_{0.3}\text{HEA}$ achieved a 94.1% anti-biofilm rate against *Pseudomonas aeruginosa* and an R_p value 20 times higher than that for Al_0HEA , effectively balancing anti-biofouling and anti-MIC properties. Specifically, the Cu element induces cell membrane damage and intracellular reactive oxygen species. Al acts as a stabilizer of the Cu-enriched phase by decreasing the potential difference between the FeCoCrNi-phase and the Cu-enriched-phase, impeding the excessive dissolution of the Cu-enriched phase while maintaining excellent anti-biofouling performance. Additionally, Al enhances the components of Al and Cr oxides in passive films, providing robust defense against corrosive species, and ultimately enhancing the MIC resistance of $\text{Al}_{0.3}\text{HEA}$. This work presents a new strategy for designing MIC-resistant materials with broad applications in marine engineering.

microbial communities.^[1] The colonization of metal surfaces by marine microorganisms is particularly concerning, as biofilm formation can induce microbiologically influenced corrosion (MIC) and biofouling,^[2] compromising structural integrity and resulting in substantial global economic losses.^[1,3] While numerous alloys such as TC4 titanium alloy,^[4] 2205 DSS,^[5] and 2304 DSS^[6] exhibit resistance to marine corrosion, they remain susceptible to marine MIC and biofouling. Consequently, the development of advanced alloys with dual anti-MIC and anti-biofouling capabilities is critically needed.

With the expansion of industrial applications, the demand for high-performance alloys has surged across various sectors.^[7] High-entropy alloys (HEAs) have emerged as a promising candidate,^[8] exhibiting exceptional properties such as high hardness,^[9] superior plasticity,^[10] enhanced wear resistance,^[11] and unique electrical and magnetic characteristics.^[12] However, studies revealed that HEAs are susceptible to corrosion, including MIC,^[13]

1. Introduction

The marine environment is exceptionally harsh on metallic materials due to its high chloride concentration and diverse

which limits their broader application in environments prone to microbial activity. To overcome this challenge, novel strategies are required to enhance their resistance to MIC and biofouling, thereby expanding their industrial applicability.^[14]

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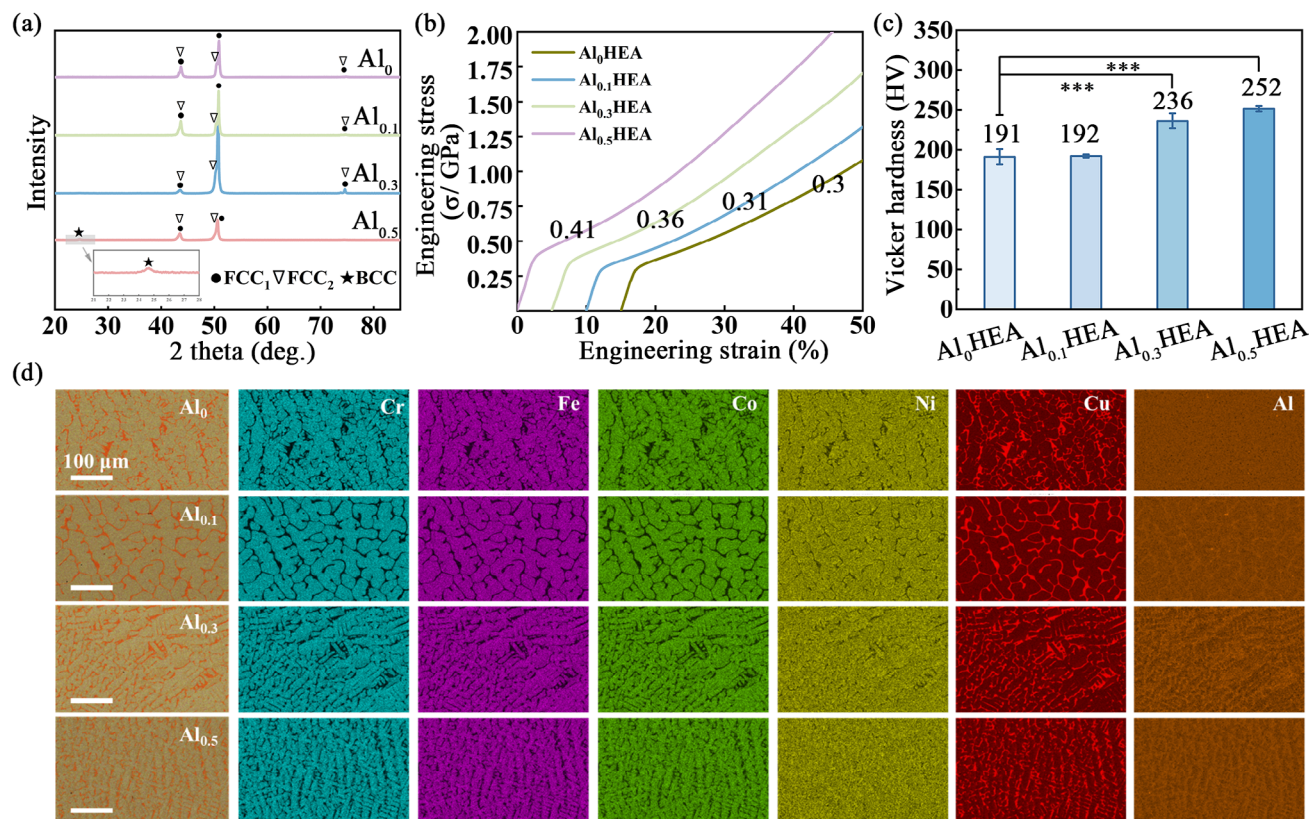


Figure 1. Microstructures and mechanical properties of Al_xHEAs (molar fractions $x = 0, 0.1, 0.3, 0.5$). a) XRD patterns. b) Engineering stress. c) Vickers hardness. d) Elemental mapping of Al, Co, Cr, Cu, Fe, and Ni elements in Al_xHEAs. The data are presented as mean values $\pm$ SD, $n = 3$, * represents $p < 0.05$.

Incorporating Cu elements into HEAs inhibits microbial growth and enhances MIC resistance.^[15] Previous studies have highlighted the antibacterial efficacy of various Cu-bearing HEAs, including CoCrCuFeNi,^[16] Fe_{1.3}CrNiCu_{0.7}Si_{0.2},^[17] Co_{0.4}FeCr_{0.9}Cu_{0.3},^[18] and CoCrFeNiCu_{0.3}.^[19] However, the high mixing enthalpy between Cu and other elements in CoCrCuFeNi-series HEAs often results in pronounced Cu segregation at dendrite interfaces,^[20] which compromises the alloy's resistance to chloride and acidic environments.^[21] As a result, optimizing the addition of Cu to enhance antibacterial properties while maintaining overall alloy performance remains a significant challenge.

Aluminum has been explored as a promising element for enhancing phase transformation,^[22] improving corrosion resistance,^[23] and strengthening the mechanical properties of HEAs.^[24] Research has demonstrated that increasing Al content reduces defect density in passive films on metal surfaces, thereby enhancing the corrosion resistance of CoNiAl_x medium-entropy alloys.^[25] Similarly, Al_xCoCrFeNi HEAs are produced through laser melting deposition with a 10 mol% Al addition, exhibiting superior corrosion resistance, attributable to beneficial alterations in the Cr₂O₃ content of the passive film and its thickness.^[26] Building on these insights, the incorporation of Al into CoCrCuFeNi-based HEAs may offer an effective approach to mitigating MIC.

To investigate the influence of Al addition on the properties of Al_xCoCrCuFeNi HEAs, four alloys with varying Al con-

tents were synthesized and analyzed in this study. This investigation provides a comprehensive evaluation of the alloys' mechanical properties, microstructures, anti-biofouling (specifically anti-microbiofouling in this study) performance, and MIC resistance. Detailed mechanistic analysis demonstrates the successful integration of mechanical strength, anti-biofouling, and corrosion resistance through the strategic modulation of elemental composition. The findings of this study aim to guide the design and production of high-performance HEAs for marine applications.

2. Results

2.1. Characterizations of Al_xHEAs

Figure 1a shows that Al₀HEA, Al_{0.1}HEA, and Al_{0.3}HEA display two face-centered cubic (FCC) phase structures. One is FCC₁ with three peaks observed at 43.7°, 51°, and 74.6°, corresponding to the (111), (200), and (220) planes, respectively. Another is FCC₂ with peaks at 43.8°, 50.2°, and 74.6°. While the Al_{0.5}HEA exhibits FCC and body-center cubic (BCC) phases, with the BCC phase appearing at 24.6°, corresponding to the (110) plane. Elemental analysis (Figures S1, Supporting Information; Figure 1d) indicates that Al_xHEAs (molar fractions $x = 0, 0.1, 0.3$) possessed a dark-gray matrix with a dendrite structure (FCC₁ phase), enriched in Co, Cr, Fe, and Ni, and a light-gray precipitate with

interdendrite structure (FCC₂ phase), composed of Al and Cu. Besides, the addition of Al leads to grain refinement in Al_xHEAs (Figure 1d). For Al_{0.5}HEA, regions enriched in Al and Ni are observed in Figure S2 (Supporting Information), consistent with the X-ray diffraction (XRD) results, confirming the presence of an Al-Ni-rich BCC phase in Al_{0.5}HEA.

The effects of Al addition on the mechanical properties of Al_xHEAs were evaluated through engineering stress and Vickers hardness measurements. Figure 1b shows that Al incorporation increased yield strength, with Al_{0.5}HEA exhibiting the highest yield strength of 0.41 GPa. The yield strength of Al_{0.3}HEA (0.36 GPa) also surpasses that of 316L SS (0.17 GPa) reported by Bernauer et al.^[27] Besides, there are no fractures under a 50% height reduction during compression testing, indicating that all Al_xHEAs demonstrate good compressive ductility.^[28] The hardness values for Al₀HEA, Al_{0.1}HEA, Al_{0.3}HEA, and Al_{0.5}HEA are 191, 192, 236, and 252 HV, respectively (Figure 1c). The results above demonstrate that the microstructure and mechanical properties of Al_xHEAs were significantly influenced by the addition of Al elements.

As the Al element tends to disperse in the Cu-enriched phase (Figure 1d), the effects of Al additions on the potential difference of Al_xHEAs were observed using the Scanning Kelvin Probe Microscopy (SKPM) model of atomic force microscope (AFM). Figure 2 presents height maps that illustrate the microstructures of Al_xHEAs with varying Al concentrations. In these maps, the dark regions correspond to the FCC₁ phase, while the light regions correspond to the FCC₂ phase. The surface potential analysis indicates that the potential differences between FCC₁ and FCC₂ phases on Al₀HEA, Al_{0.1}HEA, Al_{0.3}HEA, and Al_{0.5}HEA surfaces are 72.3 ± 7.2 mV, 54.8 ± 9.2 mV, 41.6 ± 4.0 mV, and 22.5 ± 4.0 mV, respectively. The aforementioned results demonstrate that the potential difference between the two phases was significantly reduced by doping Al in the Cu-enriched phases.

2.2. Anti-Biofouling Evaluation of Al_xHEAs

In marine environments, diverse microorganisms, including bacteria and algae, readily adhere to alloy surfaces, promoting MIC and biofouling, which ultimately compromise the longevity of the alloys. To assess their effectiveness, the anti-biofouling properties of the Al_xHEAs were evaluated first.

2.2.1. Anti-Proteins and Anti-Algae Performances of Al_xHEAs

Figure 3a shows that the highest protein concentration is recorded on the 316L stainless steel (SS) surface (0.26 ± 0.03 mg·cm⁻²) after 7 days of immersion, while the lowest is on the Al₀HEA's surface (0.11 ± 0.01 mg·cm⁻²). Additionally, compared to 316L SS, significant reductions in protein adsorption are also noted for Al_{0.3}HEA (0.17 ± 0.03 mg·cm⁻²) and Al_{0.5}HEA (0.19 ± 0.02 mg·cm⁻²), indicating that protein adsorption was inhibited on all Al_xHEAs. Subsequently, *Chlorella* sp. and *Phaeodactylum tricornutum* were used as model algae to assess the anti-algae adhesion properties of Al_xHEAs by immersing the coupons in a fresh algae suspension for 7 days under daylight conditions. In Figure 3b, the anti-algae rates of Al_{0.3}HEA against *Chlorella* sp. and *P. tricornutum* are 88 and 96%, respectively.

To further confirm the anti-algae performances of Al_xHEAs, cell respiration activity studies were conducted. As shown in Figure 3c, the Al₀HEA group exhibits the lowest respiration rates, with ≈3.4% for *Chlorella* sp. and 3% for *P. tricornutum*, while the Al_{0.3}HEA group shows 14% for *Chlorella* sp. and 5% for *P. tricornutum* compared to 316L SS, respectively. Additionally, algal biomass observations under confocal laser scanning microscope (CLSM) shows similar trends in microalgae attachment across different test alloys (Figure 3d). Notably, intense green fluorescence is detected on 316L SS, indicating abundant algal cell adhesion. In contrast, minimal green fluorescence is detected on Al_xHEAs (molar fractions $x = 0, 0.1, 0.3$) after 7 days of treatment. These findings imply that all the Al_xHEAs (molar fractions $x = 0, 0.1, 0.3$) exhibit remarkable capabilities in eliminating the algal biofilm formation by inhibiting both the attachment and the respiration of algae on alloy surfaces.

2.2.2. Anti-Biofilm Abilities of Al_xHEAs

The anti-biofilm capabilities of Al_xHEAs were further investigated since some corrosive bacteria not only give rise to biofouling but also directly induce MIC, which ultimately leads to the degradation of the alloys.^[29] The efficacy of Al_xHEAs in preventing corrosive marine *Pseudomonas aeruginosa* sessile cells compared to 316L SS was first quantitatively assessed by enumerating the number of colonies in removed biofilm samples (Figure S3, Supporting Information). After 1 day of incubation, the anti-biofilm rates of Al₀HEA, Al_{0.1}HEA, Al_{0.3}HEA, and Al_{0.5}HEA (against marine *P. aeruginosa*) are 99.99, 99.99, 92.7, and 85.4%, respectively (Figure 4a). The biofilm eradication efficiencies of Al_xHEAs after 7 days of incubation are 99.99, 99.87, 94.1, and 86.3%, respectively. These results were further validated by ATP measurements. In Figure 4b, ATP values of all groups of Al_xHEAs are lower than that of 316L SS, in particular, 325 ± 21 RLU and 2167 ± 102 RLU on the Al_{0.3}HEA and 316L SS surfaces after 7 days of immersion, respectively. Notably, although the anti-biofilm properties of Al_xHEAs decrease with increasing Al content, Al_{0.3}HEA still exhibits over 90% biofilm prevention capacity after 7 days of immersion.

CLSM images reveal the distribution of live and dead sessile cells on 316L SS and Al_xHEAs after 1, 4, and 7 days of immersion, with green fluorescence (stained with SYTO 9) representing live cells, and red fluorescence (stained with Propidium iodide) representing dead cells.^[30] The biofilm on the surface of 316L SS appears dense and robust (Figure 4c), while green fluorescence decreases significantly on Al_xHEA surfaces, particularly in lower Al content samples, indicating a diminished biofilm formation. However, this reduction becomes less pronounced with increasing Al content, suggesting a potential decline in anti-biofilm efficacy.

The inhibitory effect on the biofilm is minimized with Al addition, as more cells adhere on the surface of Al_{0.5}HEA, which is consistent with the anti-biofilm rates and ATP measurements. Scanning electron microscope (SEM) and field-emission transmission electron microscopy (TEM) images represent the change in cell morphologies after 1 day of immersion on 316L SS and Al_{0.3}HEA surfaces. Figure 4d shows that *P. aeruginosa* cells on 316L SS exhibit intact and smooth membrane structures,

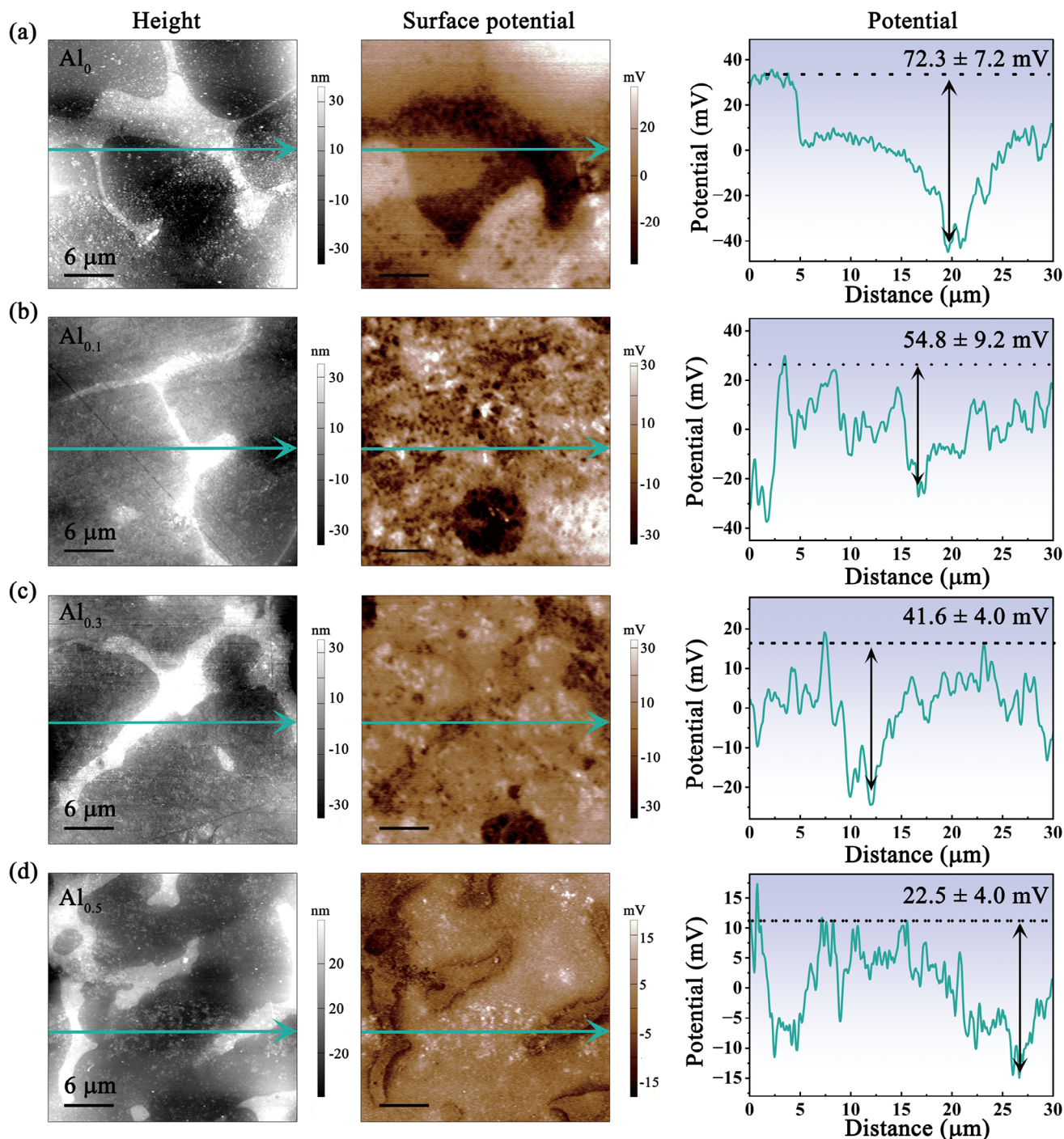


Figure 2. Surface potential analysis. Height maps, surface potential maps and surface potential profiles of the selected line scan on the Al_xHEA coupons: a) Al_0HEA , b) $\text{Al}_{0.1}\text{HEA}$, c) $\text{Al}_{0.3}\text{HEA}$, and d) $\text{Al}_{0.5}\text{HEA}$.

whereas $\text{Al}_{0.3}\text{HEA}$ caused the formation of perforations in the *P. aeruginosa* membranes.^[38] The TEM tests verify alterations in bacterial morphology and membrane integrity with or without $\text{Al}_{0.3}\text{HEA}$ treatment, confirming that $\text{Al}_{0.3}\text{HEA}$ significantly compromised the integrity of the bacterial outer membranes, ultimately leading to cell death (Figure 4e). To further investigate the anti-biofilm behavior of Al_xHEAs , reactive oxygen species (ROS)

by DCFH-DA assay in sessile cells was observed (Figure 4f). ROS, generated by antibacterial ions such as Cu ions, effectively inhibit and kill bacterial cells by damaging their membranes, proteins, and DNA. The fluorescence intensity observed on the sample surface correlates with ROS accumulation levels.^[31] Notably, minimal fluorescence (1.2%) is detected on the surface of 316L SS. In contrast, Al_0HEA exhibits a substantial ROS

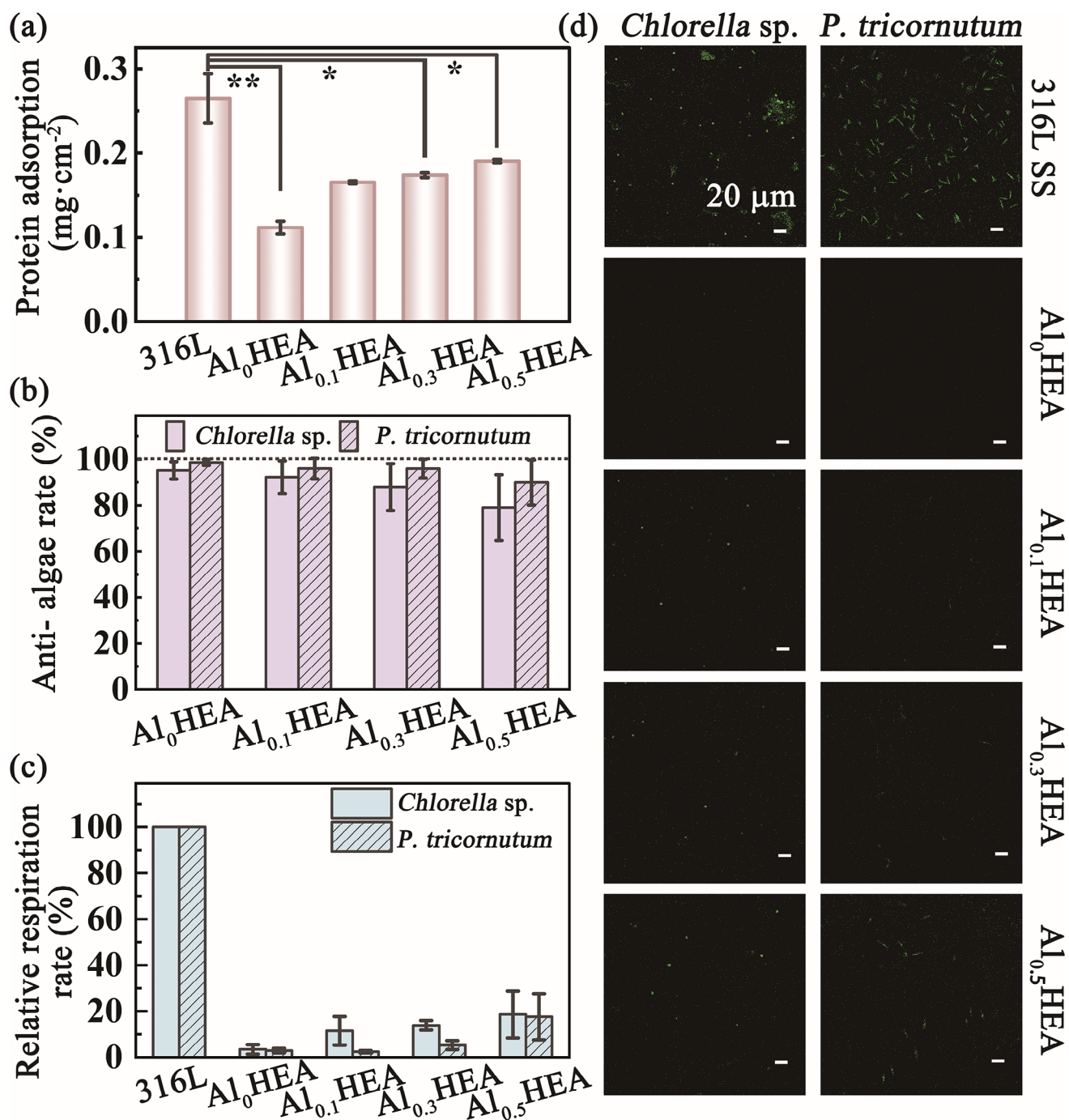


Figure 3. Anti-protein and anti-algae adhesion abilities of Al_xHEAs (molar fractions $x = 0, 0.1, 0.3, 0.5$). a) Protein adsorption on Al_xHEAs and 316L SS. b) Anti-algae rates of Al_xHEAs against *Chlorella* sp. and *P. tricornutum*. c) Relative respiration rates of algae formed on Al_xHEAs. d) CLSM images of *Chlorella* sp. and *P. tricornutum* cell mass on Al_xHEAs and 316L SS after incubating for 7 days at 20°C. The data are presented as mean values $\pm$ SD, $n = 3$, * represents $p < 0.05$.

accumulation, with fluorescence intensity reaching $\approx 17.3\%$, suggesting a significantly higher oxidative stress potential that aligns with its superior anti-biofilm performance. However, ROS levels tend to decrease with increasing Al content, with the lowest fluorescence intensity observed on the Al_{0.5}HEA surface, indicating a corresponding decline in anti-biofilm efficacy.

2.3. Anti-MIC Behaviors of Al_xHEAs

The corrosion resistance of HEAs is essential for applications in marine environments, where high chloride concentrations and corrosive microorganisms present challenges. To evaluate the MIC resistance of Al_xHEAs in marine environments, their

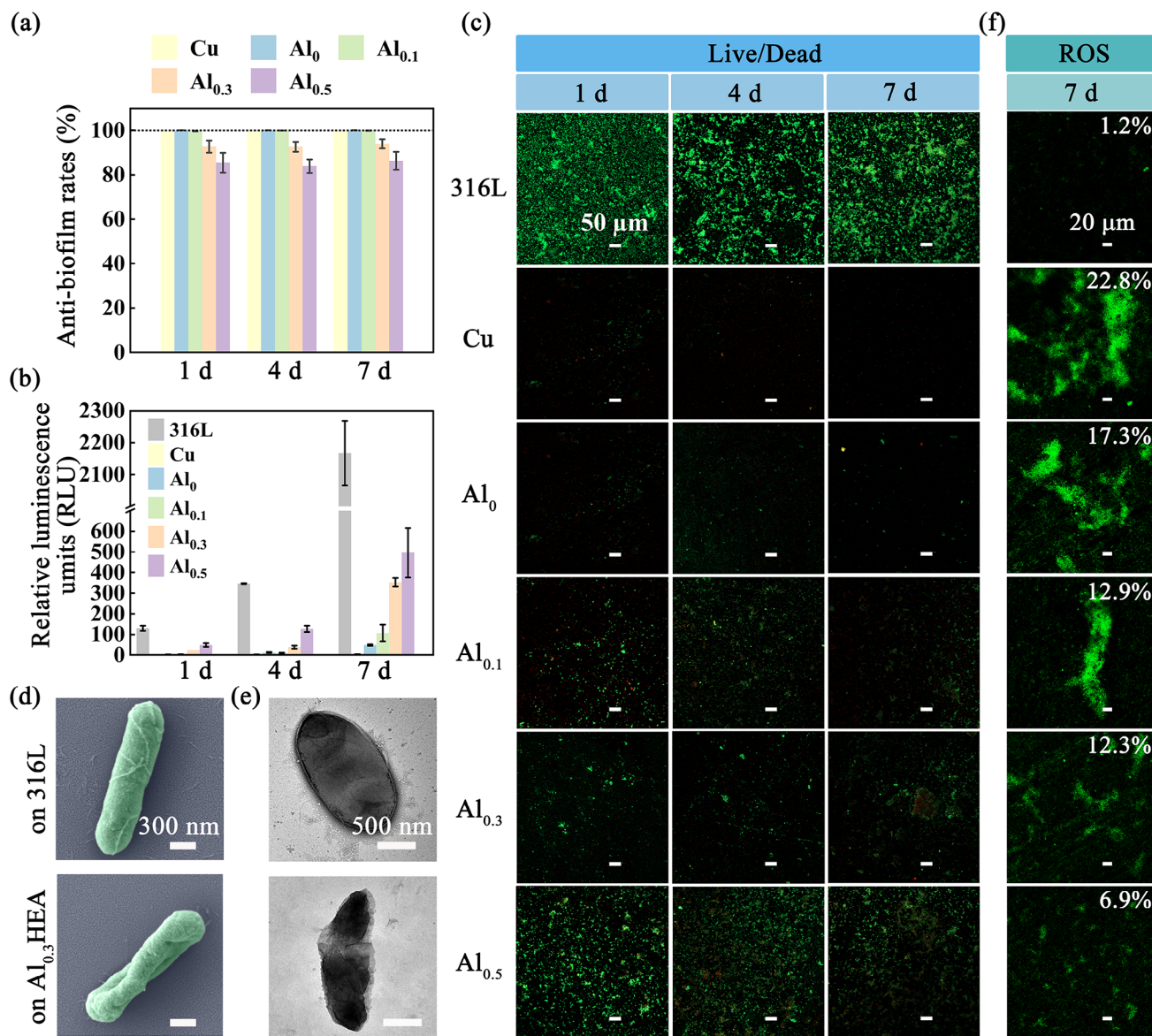


Figure 4. Anti-biofilm properties against *P. aeruginosa* on Al_xHEAs (molar fractions $x = 0, 0.1, 0.3, 0.5$) in artificial seawater. a) Anti-biofilm rates of Al_xHEAs and Cu coupons. b) ATP-bioluminescence results of sessile cells of *P. aeruginosa* after treatment by Al_xHEAs, 316L SS, and Cu coupons. c) *P. aeruginosa* biofilm observation under CLSM after incubating with Al_xHEAs, 316L SS, and Cu. d) Morphological images of *P. aeruginosa* sessile cells on 316L SS and Al_{0.3} HEA after 7 days of immersion viewed under SEM, and e) under TEM. f) ROS analysis on Al_xHEAs surfaces after 7 days of immersion under CLSM. The data are presented as mean values $\pm$ SD, $n = 3$.

corrosion performance was assessed in both abiotic and biotic artificial seawater. After 1 day of immersion, Al₀HEA has the highest $1/R_p$ values ($20.6 \pm 1.6 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$ for abiotic and $187 \pm 49 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$ for biotic), whereas Al_{0.3}HEA exhibits the lowest values ($1.4 \pm 0.1 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$ for abiotic and $2.3 \pm 1.1 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$ for biotic) in artificial seawater (Figure 5a,b), indicating that corrosion rate of Al_{0.3}HEA was the lowest.^[32] After 7 days of immersion in the presence of *P. aeruginosa*, the $1/R_p$ value for Al_{0.3}HEA ($5.8 \pm 1.4 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$) is also lower than that of 316L SS ($7.1 \pm 0.5 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$). Al_{0.1}HEA and Al_{0.5}HEA have significantly higher values, reaching up to $22.1 \pm 4.3 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$ and $32.4 \pm 4.3 \text{ M}\Omega^{-1}\cdot\text{cm}^{-2}$, respectively.^[33]

Figure 5c,d shows the potentiodynamic polarization curves of Al_xHEAs and 316L SS after 7 days of immersion in abiotic and biotic artificial seawater. The corrosion current density (i_{corr}) values obtained from the curves are presented in Figure 5e, the i_{corr} values of 316L SS in abiotic and biotic artificial seawater are $8.4 \pm 1.3 \text{ nA}\cdot\text{cm}^{-2}$ and $202 \pm 36 \text{ nA}\cdot\text{cm}^{-2}$, respectively, indicating corrosion acceleration by *P. aeruginosa*.^[34] The i_{corr} of Al_{0.3}HEA is lower than that of 316L SS in the presence of *P. aeruginosa*, registering at $157 \pm 13 \text{ nA}\cdot\text{cm}^{-2}$ after 7 days of immersion. Furthermore, the i_{corr} values for Al₀HEA, Al_{0.1}HEA, and Al_{0.5}HEA in the biotic solution measured at $1.7 \pm 0.1 \mu\text{A}\cdot\text{cm}^{-2}$, $568 \pm 49 \text{ nA}\cdot\text{cm}^{-2}$ and $325 \pm 10 \text{ nA}\cdot\text{cm}^{-2}$, respectively. These

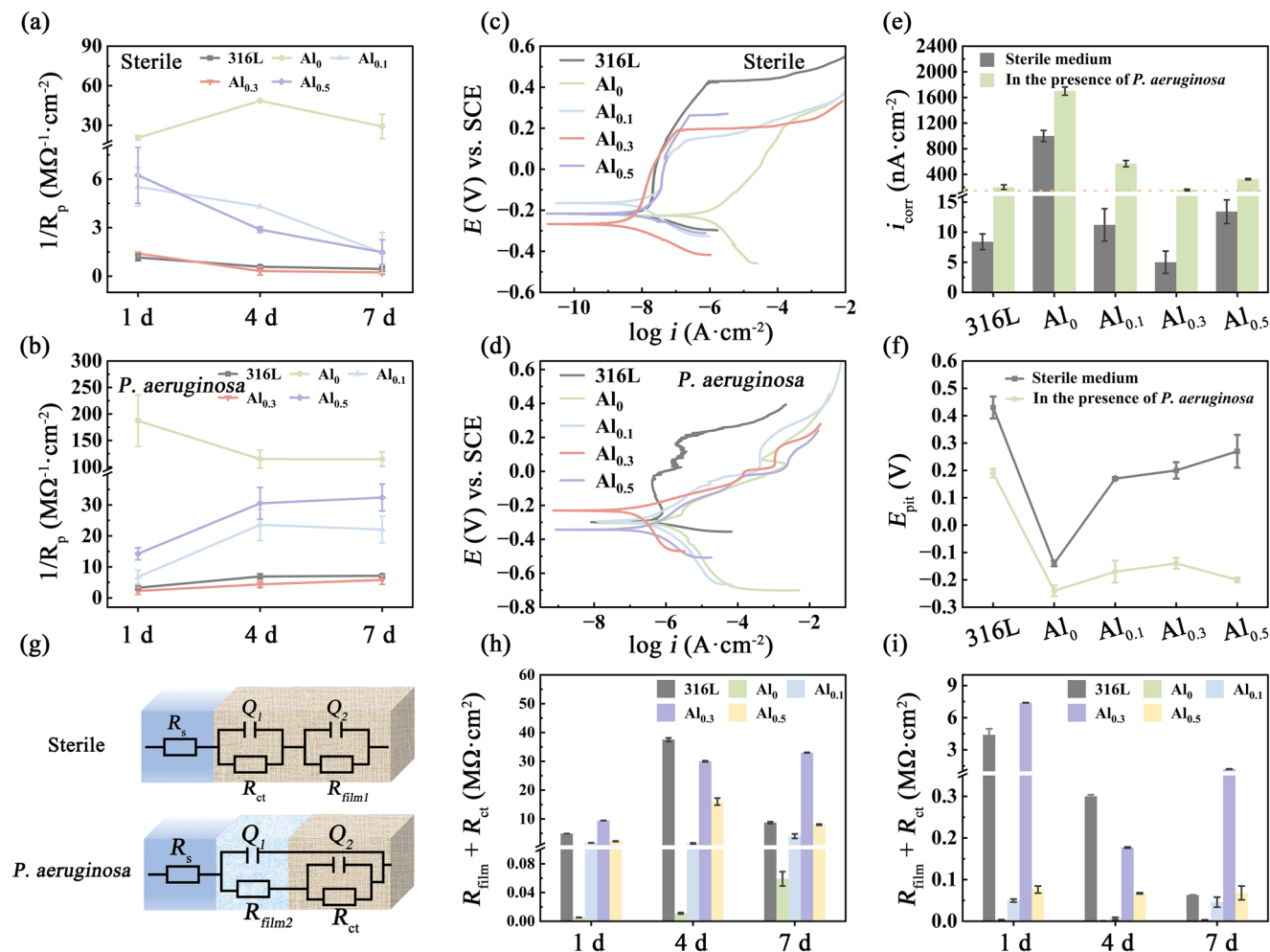


Figure 5. Electrochemical results of Al_x HEAs and 316L SS in artificial seawater with and without *P. aeruginosa*. a, b) Variations of $1/R_p$ at 1, 4, and 7 days. c, d) Potentiodynamic polarization curves at the end of 7 days of immersion. e) i_{corr} and f) E_{pit} at the end of 7 days of immersion. $R_{\text{film}} + R_{\text{ct}}$ obtained from EIS fitting data: g) equivalent circuits, h) $R_{\text{film}} + R_{\text{ct}}$ value in the abiotic medium, and i) in the presence of *P. aeruginosa*. The data are presented as mean values $\pm$ SD, $n = 3$.

results demonstrate that $\text{Al}_{0.3}$ HEA exhibits the highest MIC resistance compared with other Al_x HEAs (molar fractions $x = 0, 0.1, 0.5$) and 316L SS. The pitting potential (E_{pit}) was also analyzed to investigate the effect of Al addition on the pitting corrosion resistance of Al_x HEAs. Specifically, after 7 days in abiotic artificial seawater, the E_{pit} values for Al_0 HEA, $\text{Al}_{0.1}$ HEA, $\text{Al}_{0.3}$ HEA, and $\text{Al}_{0.5}$ HEA, are 0.15 ± 0.01 V, 0.26 ± 0.01 V, 0.35 ± 0.01 V, and 0.24 ± 0.01 V, respectively, suggesting that $\text{Al}_{0.3}$ HEA also possesses the best pitting corrosion resistance. Furthermore, the presence of *P. aeruginosa* led to a significant decrease in E_{pit} values for all alloys (Figure 5f), suggesting that the passive films are weakened by *P. aeruginosa*, which enhances uniform corrosion and reduces pitting resistance of alloys.

The electrochemical impedance spectroscopy (EIS) data were utilized to assess the influence of Al on the corrosion resistance (Figure S4, Supporting Information). The EIS data were fitted with an equivalent circuit model (Figure 5g) using the ZSimpWin software (ZsimpWin 3.60). In abiotic artificial seawater, the electrochemical behavior displayed a capacitive component in parallel with a resistive response, represented by an equivalent circuit

featuring two distinct time constants, indicative of the presence of an oxide layer on the alloy surface.^[35] In the presence of *P. aeruginosa*, biofilm formation on the coupon surface introduced an additional interfacial characteristic, with the second time constant associated with the biofilm layer/electrode interface.^[36] In the equivalent circuits, R_s , R_{ct} , R_{film} (R_{film1} and R_{film2}), Q_1 , and Q_2 are the uncompensated resistance, charge transfer resistance, film resistance, double layer constant phase element, and film constant phase elements, respectively.

After immersion for 7 days, there was a significant reduction of $R_{\text{film}} + R_{\text{ct}}$ value in 316L SS compared with biotic with abiotic artificial seawater, confirming the serious damage induced by *P. aeruginosa* (Figure 5h,i). At the same time, $\text{Al}_{0.3}$ HEA has the highest $R_{\text{film}} + R_{\text{ct}}$ value reaching up to $1.2 \times 10^6 \Omega \cdot \text{cm}^2$ after immersion in biotic artificial seawater for 7 days, whereas the values are $4.6 \times 10^4 \Omega \cdot \text{cm}^2$ for $\text{Al}_{0.1}$ HEA and $6.8 \times 10^4 \Omega \cdot \text{cm}^2$ for $\text{Al}_{0.5}$ HEA, respectively. This higher value of $\text{Al}_{0.3}$ HEA reveals greater difficulty in electron transfer between bacterial cells and the alloy surface,^[37] resulting in the enhanced MIC resistance of $\text{Al}_{0.3}$ HEA.

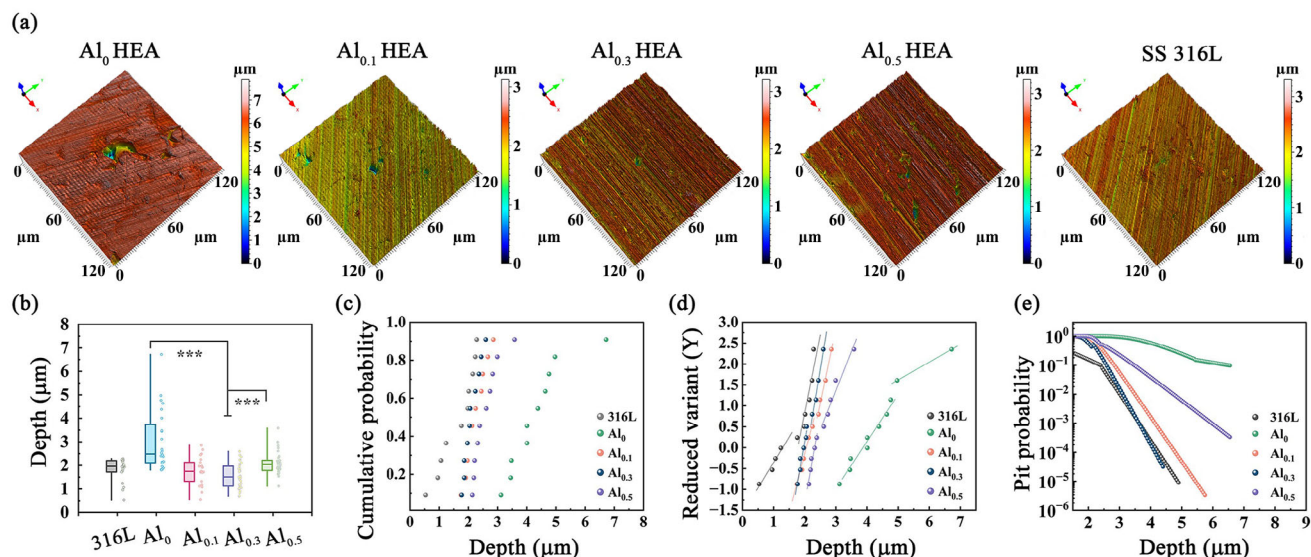


Figure 6. Pitting analysis on Al_xHEAs and 316L SS surfaces after 7 days of immersion in artificial seawater. a) Corrosion morphology after removing biofilms and corrosion products. b) The box and data plot of pit depths. Stochastic exhibition of pitting growth probability: c) cumulative probability, d) Gumbel probability plots, and e) probabilities of various pits formed on Al_xHEAs and 316L SS. The data are presented as mean values $\pm$ SD, $n = 3$, * represents $p < 0.05$.

CLSM was employed to measure corrosion pits on the surfaces of Al_xHEAs and 316L SS after 7 days of immersion in artificial seawater with *P. aeruginosa*. In **Figure 6a**, the presence of *P. aeruginosa* results in the formation of deeper pits on the Al_0HEA surface, whereas shallower pits are observed on $\text{Al}_{0.3}\text{HEA}$ surfaces. **Figure 6b** represents the pit distribution on Al_xHEAs and 316L SS, with the mean depth values for Al_0HEA , $\text{Al}_{0.1}\text{HEA}$, $\text{Al}_{0.3}\text{HEA}$, $\text{Al}_{0.5}\text{HEA}$, and 316L SS are $3.02 \pm 1.26 \mu\text{m}$, $1.78 \pm 0.59 \mu\text{m}$, $1.54 \pm 0.53 \mu\text{m}$, $2.09 \pm 0.48 \mu\text{m}$, and $1.81 \pm 0.49 \mu\text{m}$, respectively. Furthermore, **Figure 6c** demonstrates that $\text{Al}_{0.3}\text{HEA}$ has a narrower range of pit depth distribution ($1.77\text{--}2.59 \mu\text{m}$) in comparison to other Al_xHEAs (molar fractions $x = 0, 0.1, 0.5$). **Figure 6d** shows the fitting plots of pit depth from the Gumbel distribution.^[38] The pitting cumulative probabilities of Al_xHEAs and 316L SS in the presence of *P. aeruginosa* are illustrated in **Figure 6e**, which is estimated by using the ten biggest pit depths on the coupon surfaces (three replicate coupons). Similar to the results above, $\text{Al}_{0.3}\text{HEA}$ is less likely to develop deeper pits than Al_0HEA , $\text{Al}_{0.1}\text{HEA}$, and $\text{Al}_{0.5}\text{HEA}$.

2.4. Passive Film Analysis

The primary role of passive films on metal surfaces during the corrosion process is to trap metal ions underneath rendering further oxidation thermodynamically unfavorable.^[39] To evaluate the protective capacities of passive films formed on Al_xHEAs (molar fractions $x = 0.1, 0.3, 0.5$), the compositions of passive films were examined. **Figures S5–S8** (Supporting Information) illustrate the multiplex spectra of the tested alloys by X-ray photoelectron spectroscopy (XPS), exhibiting the characteristic peaks of O 1s, Al 2p, Cu 2p, and Cr 2p_{3/2} after 15 s of sputtering. The specific component ratios of Al, Cr, Cu, and O were calculated. Shown in **Figure 7a**, with the addition of Al from 0.1 to

0.3, the ratio of $\text{Cr}^{3+}/\text{Cr}^{\text{m}}$ and $\text{Al}^{3+}/\text{Al}^{\text{m}}$ (m : metallic states) in the abiotic medium increases, whereas $\text{Cu}^{2+}/\text{Cu}^{\text{m}}$ and $\text{OH}^-/\text{O}^{2-}$ decrease, respectively. Nonetheless, only the ratio of $\text{Al}^{3+}/\text{Al}^{\text{m}}$ and $\text{Cu}^{2+}/\text{Cu}^{\text{m}}$ show the same trend when Al contents increase from 0.3 to 0.5, indicating increased contents of Al-hydroxide and Cu-hydroxide in the passive film of $\text{Al}_{0.5}\text{HEA}$. In the presence of *P. aeruginosa*, the relative amounts of Al/Cr compounds in the passive films of all Al_xHEAs decrease, while Cu oxides and hydroxides increase, suggesting that the protective quality of the passive films is compromised by *P. aeruginosa*. Notably, $\text{Al}_{0.3}\text{HEA}$ exhibits the most protective passive film among the Al_xHEAs in both abiotic and biotic solutions, as it contains higher levels of Al/Cr compounds and lower concentrations of Cu compounds.

To observe the corrosion signs on the Al_0HEA and $\text{Al}_{0.3}\text{HEA}$ coupons, all the coupons were etched with 2% (v/v) nitric alcohol gently and then observed under SEM after 7 days of immersion. In **Figures S9** (Supporting Information) and **Figure 7b**, significant selective corrosion signs are observed on the Al_0HEA coupons (in both abiotic and biotic artificial seawater), which are concentrated and dispersed on the Cu-enrich phase (FCC_2). The $\text{Al}_{0.3}\text{HEA}$ demonstrates resilience against both seawater and microbial attacks, with the Cu-enrich phase (FCC_2) remaining relatively intact even in the presence of *P. aeruginosa*. These results suggest that the regions enriched with Cu, and the absence of Cr were preferentially targeted by corrosive agents. However, such a risk was effectively minimized when the Al content reached up to 0.3 (molar fraction). The concentrations of Cu ions released from Al_xHEAs after immersion in artificial seawater for 7 days were quantified. In **Figure 7c**, $28 \mu\text{g}\cdot\text{cm}^{-2}$ of Cu ions leaching from Al_0HEA , and $7.5 \mu\text{g}\cdot\text{cm}^{-2}$ from $\text{Al}_{0.3}\text{HEA}$ after immersion in abiotic seawater for 7 days. Furthermore, the presence of *P. aeruginosa* increased the Cu ions concentration, resulting in $156 \mu\text{g}\cdot\text{cm}^{-2}$ from Al_0HEA , and $66.6 \mu\text{g}\cdot\text{cm}^{-2}$ from $\text{Al}_{0.3}\text{HEA}$.

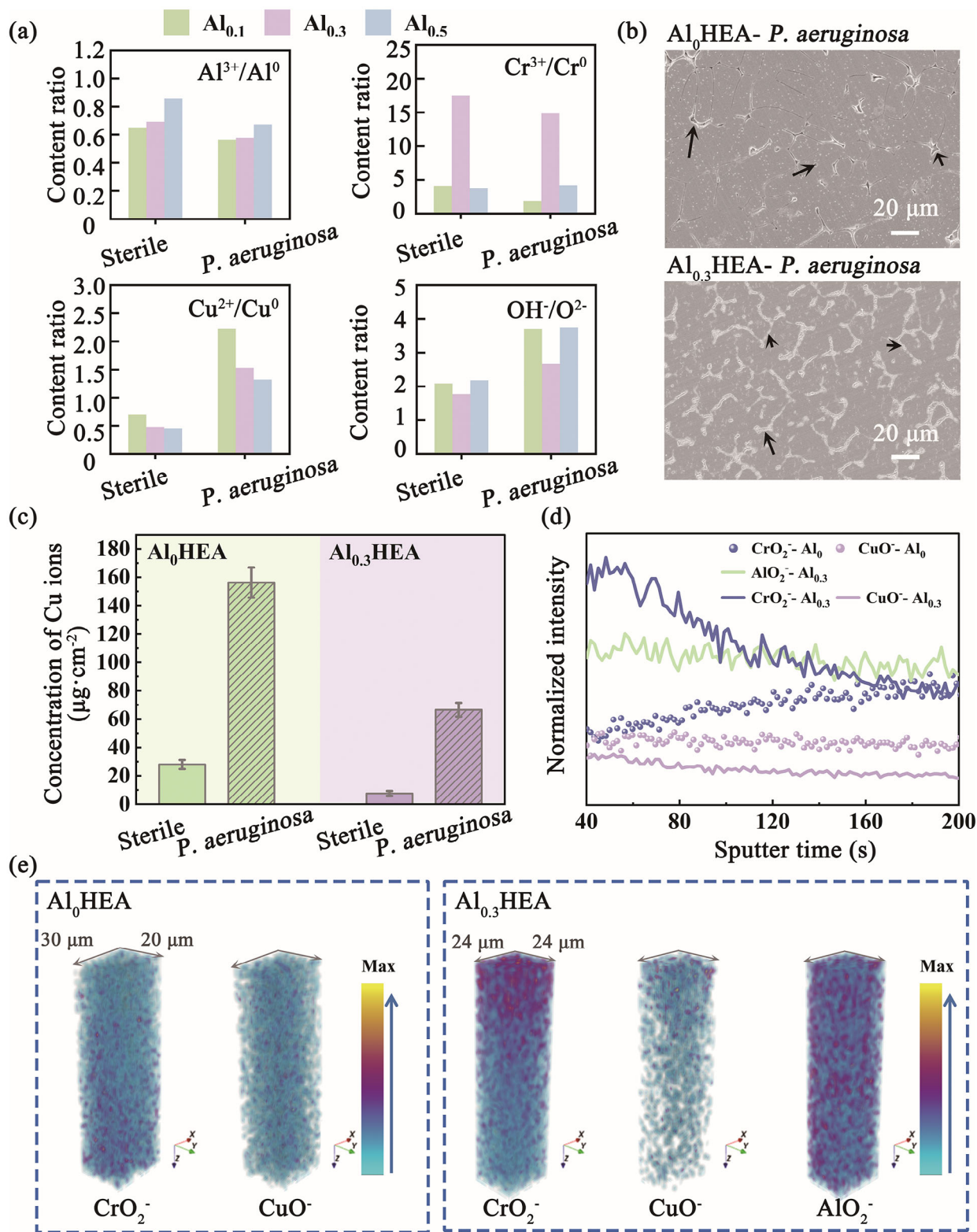


Figure 7. Analysis of passive films formed on Al_xHEAs (molar fractions $x = 0, 0.1, 0.3, 0.5$). a) The ratio of Al³⁺/Al^m, Cr³⁺/Cr^m, Cu²⁺/Cu^m, and OH⁻/O²⁻ in the passive film of Al_xHEAs ($x = 0, 0.1, 0.3, 0.5$) measured by XPS. b) Corrosion morphology of the Al_xHEAs ($x = 0, 0.3$) in the presence of *P. aeruginosa* after incubation for 7 days. c) The concentrations of Cu ions released from Al_xHEAs ($x = 0, 0.3$) after incubation for 7 days in artificial seawater (abiotic and biotic). d) TOF-SIMS depth profiles of the anions in the passive films layer on Al_xHEAs ($x = 0, 0.3$) after 7 days of incubation with *P. aeruginosa*, and e) 3D tomography results of passive films for Al_xHEAs ($x = 0, 0.3$) after 7 days of incubation with *P. aeruginosa*. The data are presented as mean values $\pm$ SD, $n = 3$.

The time of flight secondary ion mass spectrometry (TOF-SIMS) analysis was employed to visualize the effect of Al addition on the composition of the passive films on the Al_xHEAs surfaces, in which only secondary anions were recorded and compared. Figure 7d represents a typical anion SIMS depth profile, in which the scatter represents Al_0HEA and the curve represents $\text{Al}_{0.3}\text{HEA}$. Notable differences in ion intensities can be observed between Al_0HEA and $\text{Al}_{0.3}\text{HEA}$, such as AlO_2^- , CuO^- , and CrO_2^- . The normalized intensity of CuO^- in Al_0HEA is higher than that of $\text{Al}_{0.3}\text{HEA}$, whereas the intensity of CrO_2^- was lower than that of $\text{Al}_{0.3}\text{HEA}$. Besides, the normalized intensities of CrO_2^- and AlO_2^- in $\text{Al}_{0.3}\text{HEA}$ are significantly higher than that of CuO^- . The 3D tomography renderings for Al_xHEAs ($x = 0, 0.3$) are shown in Figure 7e. CuO^- is uniformly distributed in Al_0HEA , while CrO_2^- grows progressively with sputtering time. On the other hand, AlO_2^- is spread throughout the entire passive film of $\text{Al}_{0.3}\text{HEA}$, while CrO_2^- is scattered on the top of the film as shown in Figure 7e. The significant improvement in MIC resistance obtained by the Al addition was primarily attributed to its regulation of the Cr oxide and the effective shielding of Cu compounds within the passive film.

3. Discussion

3.1. Enhanced Mechanical Properties of Al_xHEAs

Microstructure analysis of Al_xHEAs shows typical dendrite and interdendritic structures, identified as two distinct FCC structures by XRD (Figure 1a). Elemental analysis confirms that the interdendritic regions, with smaller volume fractions, are enriched with Cu, while the dendritic regions, with larger volume fractions, consist of multi-principal elements (CoCrFeNi) (Figure 1d).^[40] This segregation occurs due to the low binding energy between Cu and Co, Cr, Fe, and Ni, causing Cu to preferentially concentrate in the solidified regions.^[41] Additionally, the addition of Al, with its larger atomic radius, increases lattice-distortion energy, potentially leading to a transformation in crystal structure from FCC to BCC.^[42] As a result, the BCC phase appears identified as Al, Ni-enriched when the Al content is up to 0.5 (molar fractions) (Figure S2, Supporting Information), resulting from the decomposition of the FCC phase.^[43] In fact, with the addition of Al, the amount of FCC/BCC eutectic phases in the Al_xHEAs gradually decreased, and the BCC phase might gradually replace the FCC phase and become the dominant one.^[41]

It is well established that the strength and toughness of alloys are closely tied to their microstructures, and the observed microstructural changes in Al_xHEAs with Al addition suggest an enhancement in mechanical properties. Consequently, the highest value of yield strength and hardness appears in $\text{Al}_{0.5}\text{HEA}$, which is up to 0.41 GPa (Figure 1b) and 252 HV (Figure 1c), respectively. This improvement in mechanical properties can be attributed to the refined grain structure, as illustrated in Figure 1d. Previous studies have shown that the addition of Al refines the grain structure, increasing grain boundary density and thereby enhancing yield strength and hardness by impeding dislocation motion in AlCoCrFeNiTi HEA.^[44] Additionally, another plausible reason is that the lattice distortion induced the quantity of

Al-Ni intermetallic, all of which significantly increases the hardness of HEA.^[45] Thus, the strengthening of HEAs through Al addition can be attributed to solid solution strengthening, dispersion strengthening, and nano-phase strengthening.^[44] Notably, the enhanced strength and hardness did not sacrifice compressive ductility, due to the excellent plasticity of the FCC phases,^[46] Al_xHEAs possesses superior tensile properties because none fractured under a 50% height reduction during compression testing (Figure 1b).

3.2. Sustainable Biofouling Mitigation Properties Benefiting from Cu

Biofouling, driven by the adhesion and colonization of microorganisms such as bacteria and algae, is a persistent and highly destructive process that causes irreversible surface deterioration and severely compromises material performance.^[47] Designing alloys with inherent anti-biofilm and anti-algae properties offers a more convenient and effective solution to mitigate biofouling compared to external physical interventions or coatings. Therefore, Cu was introduced in this work, aiming to obtain superior anti-biofouling properties. Compared with 316L SS ($0.26 \pm 0.03 \text{ mg}\cdot\text{cm}^{-2}$), a significantly lower value of protein adhesion on the surface of the Al_xHEAs , with the lowest concentration on the Al_0HEA surface ($0.11 \pm 0.01 \text{ mg}\cdot\text{cm}^{-2}$) (Figure 3a), indicate the interference with the initial microbial adhesion. These results imply the potential anti-biofouling capacity of the Al_0HEA by reducing the opportunity for microbial colonization. Despite the attenuation, the $\text{Al}_{0.3}\text{HEA}$ also exhibits excellent performance in terms of resistance to protein adhesion, registering a value of $0.17 \pm 0.03 \text{ mg}\cdot\text{cm}^{-2}$.

The anti-biofilm and anti-algae results demonstrate a consistent trend. Specifically, all Al_xHEAs exhibit significant anti-algae and anti-biofilm properties, with Al_0HEA showing the highest efficacy. However, as the Al content increases, both anti-biofilm and anti-algae performances decline. Despite this, $\text{Al}_{0.3}\text{HEA}$ retains notable anti-biofouling capabilities, achieving values of 94.1% against *P. aeruginosa* (Figure 4a), 88% against *Chlorella* sp. (Figure 3b), and 96% against *P. tricornutum* after 7 days of immersion.

The superior anti-biofouling performances of Al_xHEAs mainly contributed to Cu. First, the Cu element is prone to precipitate and form separate phase structures. Thus, a potential difference between the dendrite (FCC₁, Fe, Ni, Cr, Co-enriched) and interdendritic (FCC₂, Cu-enriched) structures, readily forms a galvanic couple and triggers short-range electron transfer.^[48] The resultant electrons may disrupt membrane structures, and interfere with gene transcription and translation, thus inhibiting microbial activities.^[48] This inhibitory effect on microbes diminishes as the Al content increases since the addition of Al lessens the potential difference between these two phases (Figure 2). Another factor is that Cu, an effective anti-fouling agent,^[47] enables Cu ions released from the alloy to exhibit potent anti-biofouling activities. Positively charged Cu ions readily bind to the negatively charged cell membranes of bacteria and algae via electrostatic interactions,^[49] disrupting membrane integrity, affecting cellular physiological functions, and leading to cell apoptosis.^[50] Additionally, the Cu ions that enter the cells

can damage intracellular protein structures, inhibiting microbial growth and reproduction.^[51] Cu ions can also catalyze the generation of ROS in multiple ways, causing widespread damage to various biomolecules within microbial cells.^[52] As shown in Figure 4f, compared with 316L SS, a significant amount of ROS fluorescence signals was captured on the Al_xHEAs surfaces, indicating that the alloy induced substantial ROS production, causing microbial oxidative stress and thus achieving excellent anti-biofouling performances. The ROS intensity decreases to 6.9% with an Al content of 0.5, correlating with reduced Cu ions release from Al_{0.5}HEA, implying weakened anti-biofilm properties. This occurs because Al tends to aggregate in the Cu-enriched phase, stabilizing it and further suppressing Cu ion release. These findings collectively demonstrate that Al_xHEAs (molar fraction $x = 0, 0.1, 0.3$) are effective for biofilm eradication, offering significant potential for long-term biofouling and MIC inhibition.

3.3. Al Addition Stabilizes Passive Film and Enhances MIC Resistance

The value of $1/R_p$ in Al_{0.3}HEA is almost 20 times lower than Al₀HEA after incubation in biotic artificial seawater for 7 days (Figure 5b), demonstrating that Al addition effectively enhances the MIC resistance of Al_{0.3}HEAs.^[53] However, the inhibition of Al on MIC was not always positively associated with the Al content. After 7 days of immersion in biotic artificial seawater, i_{corr} increases from $157 \pm 13 \text{ nA}\cdot\text{cm}^{-2}$ to $325 \pm 10 \text{ nA}\cdot\text{cm}^{-2}$, when Al contents vary from 0.3 to 0.5 (molar fractions) (Figure 5e). The analysis of microbial-induced pitting corrosion aligns with the electrochemical test results described above. In Figure 6, only small and shallow corrosion pits appear on the Al_{0.3}HEA surfaces, demonstrating the best anti-pitting corrosion performance among Al_xHEAs.^[54]

For deciphering the mechanism of remarkable MIC resistance in Al_{0.3}HEA, its outstanding biofilm removal performance was initially emphasized, as biofilm formation by microorganisms on metal surfaces is a critical factor in MIC. However, the results that $x = 0.3$ (molar fraction) is the optimal Al content for MIC resistance (rather than Al₀HEA) demonstrates that the MIC resistance of Al_{0.3}HEA originated from not only the biofilm eradication capabilities but also the synergistic interactions of multiple alloying elements.

The electrochemical theory for alloys suggests that a significant potential difference between neighboring phases might promote micro-galvanic corrosion.^[48] The severity of micro-galvanic corrosion increases with a higher potential difference between the two phases.^[55] The potential difference between the two phases in Al₀HEA (Figure 2) could generate electrons and contribute to antibacterial properties. However, this simultaneously induces micro-galvanic corrosion of Al_xHEAs in a corrosive environment. The Cu-enriched phase (FCC₂) has a lower potential than the matrix phase (FCC₁), thus preferentially dissolving as the anode and exhibiting visible selective corrosion signs on Al₀HEA after 7 days of immersion (Figure 7b). Remarkably, the Al element tends to aggregate more readily in the Cu-enriched phase, decreasing the potential difference and inhibiting micro-galvanic corrosion. As a result, the morphology of the Al-Cu phase in Al_{0.3}HEA is relatively more intact than the Al₀HEA

(Figures S9, Supporting Information; Figure 7b). Despite the further decreased potential difference in Al_{0.5}HEA, the BCC phase (Al-Ni-enriched and Cr-depleted) might serve as a new corrosion-prone locus, leading to the expansion of corrosion from pitting to selective dissolution,^[22] culminating in a decrease in the MIC resistance of the Al_{0.5}HEA. Thus, the investigation into the optimal content of Al is important for the modulation of the microstructure of CoCrCuFeNi HEAs for marine applications.

The optimization of alloy microstructures by adding Al is conducive to the formation of a uniform and stable passive film. Thermodynamically, the formation of the stable passive film leads to a positive shift of an alloy's electrode potential, thereby diminishing the propensity for metal oxidation. To elucidate the mechanisms underlying the MIC resistance of Al_{0.3}HEA, XPS, and TOF-SIMS were performed to investigate the passive films. XPS results show that the ratio of $\text{Cr}^{3+}/\text{Cr}^m$, $\text{Al}^{3+}/\text{Al}^m$, and $\text{O}^{2-}/\text{OH}^-$ in the passive film on Al_{0.3}HEA are higher than those in Al_{0.1}HEA and Al_{0.5}HEA (Figure 7a), indicating enrichment of Cr oxide and Al oxide in the passive films. TOF-SIMS analysis further highlights the addition of Al in favor of the distributions of AlO_2^- and CrO_2^- , and normalized intensities of AlO_2^- and CrO_2^- are significantly higher than Cu oxide (Figure 7d). The TOF-SIMS 3D tomography results reveal no spatial correlation between Cu oxide and Al oxide (Figure 7e), in which Al oxide is uniformly dispersed within the Cu-enriched regions, effectively protecting the Cu oxide. The elevation of AlO_2^- and CrO_2^- contents enhanced the stability of the passive film formed on Al_{0.3}HEA, effectively mitigating the selective corrosion on weak regions of passive films in the marine environment induced by chloride and microbes, and thus enhancing the MIC resistance.^[56]

To validate our hypothesis, the inductively coupled plasma optical emission spectrometer (ICP-OES) was conducted to measure the concentration of Cu ions released from Al₀HEA and Al_{0.3}HEA. After 7 days of immersion in abiotic artificial seawater, Al₀HEA released nearly four times more Cu ions than Al_{0.3}HEA (Figure 7c), exceeding the effect of the original Cu content (Table S1, Supporting Information). This result supports the idea that the Cu-enriched phase in Al_{0.3}HEA is less susceptible to dissolution in corrosive environments. These findings emphasize the critical role of Al addition in forming passive films on Al_{0.3}HEA, which are enriched with Al-Cr oxides and resist dissolution.^[57] A qualitative comparison with existing middle/high-entropy alloys, based on key marine application metrics, highlights the potential of Al_{0.3}HEA as an effective and versatile solution for anti-biofouling and robust anti-MIC performance (Figure 8).

To summarise, the exceptional anti-MIC properties of Al_{0.3}HEA had a strong relationship with excellent anti-biofilm properties and protective passive films on the surface. The synergistic effects of Al-Cu elements confer MIC resistance to Al_{0.3}HEA, as shown in Figure 9. Specifically, the gradually sustained Cu ions released from Al_{0.3}HEA could act as anti-biofilm agents, effectively inducing the rupture of bacterial cell membranes and hindering the adhesion of algae on the coupon surfaces. Importantly, the superior biofilm prevention performance of Al_{0.3}HEA does not compromise its corrosion resistance. In addition to accumulating in the Cu-enriched phase, Al is oxidized and integrated with Cu oxide within the

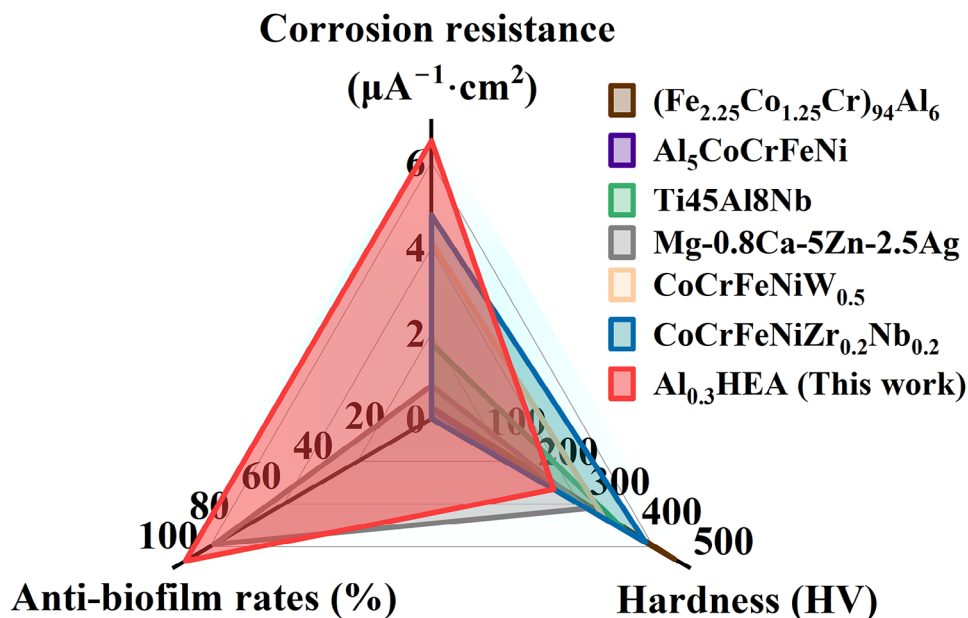


Figure 8. Corrosion resistance versus anti-biofilm rate versus hardness of $\text{Al}_{0.3}\text{HEA}$ compared with other reported middle/high-entropy alloys. The performance parameter of the benchmark alloys including cast ($\text{Fe}_{2.25}\text{Co}_{1.25}\text{Cr}$) $_{94}\text{Al}_6$,^[58] $\text{Al}_5\text{CoCrFeNi}$,^[26] Ti45Al8Nb,^[59] Mg-0.8Ca-5Zn-2.5Ag,^[60] $\text{CoCrFeNiW}_{0.5}$,^[61] and $\text{CoCrFeNiZr}_{0.2}\text{Nb}_{0.2}$,^[62] are taken from literature.

passive film, enhancing Cu oxide passivation and enabling the controlled release of Cu ions. This integration, combined with the presence of Cr as an alloying element, results in the formation of a robust and defect-resistant passive film that inhibits further oxidation and corrosion by blocking ionic transport.

4. Conclusion

In this work, a series of $\text{Al}_x\text{CoCrCuFeNi}$ high-entropy alloys (molar fraction $x = 0, 0.1, 0.3, 0.5$) were successfully designed, and their marine anti-biofouling and anti-MIC performances were systematically evaluated. The results highlight that the precise tailoring of chemical compositions enabled $\text{Al}_{0.3}\text{HEA}$ to achieve not only remarkable mechanical enhancement but also exceptional biofouling and MIC resistance. As a result, $\text{Al}_{0.3}\text{HEA}$ achieved a 94.1% anti-biofilm rate after 7 days against aerobic *P. aeruginosa* and showed a 20-fold increase in R_p value compared to Al_0HEA . This performance was attributed to the synergistic effects of Cu and Al elements, in which Cu element disrupted biofilms through catalytic actions, while Al stabilized the Cu-enriched phase, preventing excessive dissolution. The incorporation of Al also promoted the formation of protective Al oxide and Cr oxide layers within the passive films, thereby enhancing the alloy's resistance to abiotic corrosion and MIC. This study establishes a paradigm for Al/Cu-bearing HEAs, bridging the gap between homogeneous elemental design and enhanced functional performances in complex biological and corrosive marine environments. The insights gained from this study provide useful guidance for the design of metal alloys with tailored, integrated properties for applications in biologically complex marine environments.

5. Experimental Section

Preparation and Characterizations of Al_xHEAs : $\text{Al}_x\text{CoCrCuFeNi}$ high-entropy alloys (molar fractions $x = 0, 0.1, 0.3, 0.5$, denoted as Al_0 , $\text{Al}_{0.1}$, $\text{Al}_{0.3}$, $\text{Al}_{0.5}$, respectively in Al_xHEAs) were prepared from pure elements (Co, Cr, Cu, Ag, Ni: 99.9 wt%, and Fe: 99.5 wt%). The raw elements were alloyed in a copper crucible in a vacuum arc melting furnace.^[63] The furnace chamber was first evacuated to 6×10^{-3} Pa and then backfilled with high-purity argon gas (99.99% by volume) to 0.05 MPa. Raw metals were melted in a vacuum arc furnace for ≈ 45 s. The alloy buttons were turned upside down, and each alloy ingot was remelted at least 6 times to achieve chemical and microstructural homogeneity. After melting, the buttons were cross-sectionally cut into small pieces according to the needs of the different experiments. The elemental contents (wt%) of the obtained Al_xHEAs are presented in Table S1 (Supporting Information).

Tensile tests were performed on a testing machine (Instron 5569, Instron, USA) with a strain rate of $1 \times 10^{-3} \text{ s}^{-1}$ at room temperature (25°C). All metal samples were machined with a gauge length of 10.0 mm, a width of 2.0 mm, and a thickness of 2.0 mm according to the International Standards 2002. A full-range extensometer (TRViewX, Shimadzu, Japan) was used for strain measurements. Hardness measurements were made using an Amicro-Vickers hardness tester (HM-115, MITUTOYO, Japan) under a load of 1000 g with a loading time of 15 s. Measurements were replicated at least 7 points for reproducibility of each specimen group.

XRD was performed on Smart Lab9 kW (Rigaku, Japan) with a Cu target, a scanning rate of $4^\circ/\text{min}$, and a scanning range of $20^\circ - 90^\circ$. Microstructures and compositions of the alloys were analyzed using a SEM (EVO 10, ZEISS, Germany) equipped with an X-ray energy dispersive spectrometer (EDS). Electron backscattered diffraction (EBSD) was performed on a Zeiss Sigma 300 instrument with a high-resolution detector (Oxford C-Nano, UK). The surface potential was detected using an AFM (Cypher ES, Asylum Research, Oxford Instrument, UK), and the corresponding surface potential was tested using the SKPM model.^[64]

Anti-Biofouling Tests: The adsorption amount of protein on different surfaces was tested using the BCA Protein Content Assay Kit (Beijing Biotek Science & Technology Co., Ltd., China). To evaluate the anti-algae properties of Al_xHEAs , the *Chlorella* sp. and *P. tricornutum*, obtained from Guangyu Biological Technology Co., Ltd., were used as the experimental

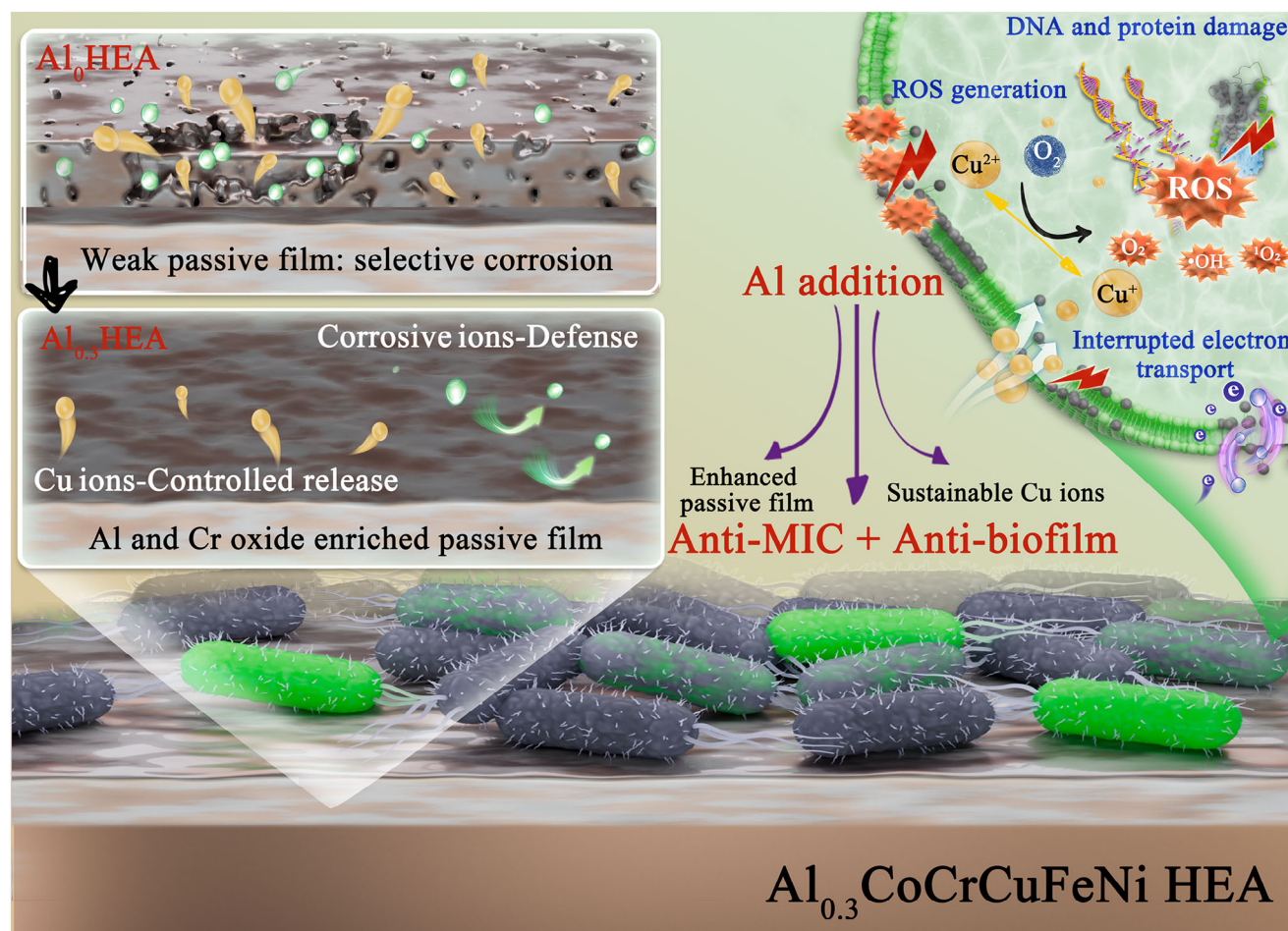


Figure 9. Schematic diagram of anti-biofilm and anti-MIC mechanisms of $\text{Al}_{0.3}\text{CoCrCuFeNi}$ HEA.

algae species. The *Chlorella* sp. and *P. tricornutum* were cultured in F/2 medium and F/2 + Si medium with visible-LED light irradiation (3000 Lux) at 20°C and 30 for 7 days (Table S2, Supporting Information). The algae broth was collected and diluted with fresh F/2 culture medium to $10^5 \text{ CFU} \cdot \text{mL}^{-1}$ for subsequent experiments, and the three testing coupons were immersed in 30 mL diluted algae solution for 7 days. The culture vessels were capped with sterile sealing films. After that, the coupons were washed with pH 7.4 phosphate buffer saline solution (PBS) ($8 \text{ g} \cdot \text{L}^{-1}$ NaCl, $0.2 \text{ g} \cdot \text{L}^{-1}$ KCl, $1.44 \text{ g} \cdot \text{L}^{-1}$ Na_2HPO_4 , and $0.24 \text{ g} \cdot \text{L}^{-1}$ KH_2PO_4) gently to remove the planktonic cells. Then, the algal biofilm cells were resuspended in 1 mL of PBS. Subsequently, the adhesion level of the algae was characterized by the absorbance at $\lambda = 440 \text{ nm}$. The anti-algae rates were calculated by the following Equation 1:

$$\text{Anti-algae rate (\%)} = (A_{\text{ctrl}} - A_{\text{anti}}) / A_{\text{ctrl}} \times 100 \quad (1)$$

where A_{ctrl} represents the absorbance at 440 nm of algal cells on the 316L SS coupon surface (control), and A_{anti} the absorbance at 440 nm of algal cells on the Al_xHEAs coupon surface.

The adhesion of algae biofilm cells was directly observed through a CLSM (LSM 900, ZEISS, Germany). To evaluate the cell respiration activity, 1 mL algal biofilm cell mixture was collected and 2,3,5-triphenyl tetrazolium chloride (TTC) (1% w/v) was added. After incubation for 7 days, the absorbance at 485 nm was regarded as an indicator of corresponding algal respiration activity.^[65]

Gram-negative *P. aeruginosa aeruginosa* (1A00099) were purchased from the Marine Culture Collection China (MCCC) and were inoculated into 2216E medium at 37°C for 20 h, and then the bacterial pellet was suspended in the artificial seawater until the OD585 values reached ≈ 0.3 . The components of artificial seawater were based on ref. [14] (Table S3, Supporting Information).

Anti-biofilm properties of Al_xHEAs were compared with traditional 316L SS and Cu, each coupon was ground from 400 grits to 1000 grits and washed with anhydrous ethyl alcohol. Anti-biofilm testing was briefly as follows. First, bacterial cultures were diluted to $10^7 \text{ CFU} \cdot \text{mL}^{-1}$ with abiotic seawater. Then, 30 mL artificial seawater (with $10^7 \text{ CFU} \cdot \text{mL}^{-1}$ of *P. aeruginosa*) and three parallel samples were added into a conical flask, incubating for 1, 4, and 7 days, respectively. After that, each coupon was washed with PBS twice to remove planktonic cells. Then, biofilm cells were diluted using PBS, and 100 μL of algal cell suspension was spread on the 2216E agar plate for incubation overnight. The anti-biofilm properties of Al_xHEAs were evaluated by counting the number of cells on the solid medium after 20 h of incubation at 30 °C. The anti-biofilm rates were calculated by the following Equation 2:

$$\text{Anti-biofilm rate (\%)} = (A_{\text{ctrl}} - A_{\text{anti}}) / A_{\text{ctrl}} \times 100 \quad (2)$$

where A_{ctrl} represents the total number of sessile cells on the 316L SS coupon surface (control), and A_{anti} the total number of sessile cells on the pure Cu or Al_xHEAs coupon surface.

The relative luminescence units (RLU) value of the sample surfaces was measured by Adenosine-5'-triphosphate (ATP) bioluminescence swabs. ATP swabs were acquired from Tian Long Technology Co., Ltd. (Xi'an, China). The adhesion of bacterial biofilm cells was directly observed through CLSM using the LIVE/DEAD BacLight Bacterial Viability Kit (Thermo Fisher Scientific, USA). The generation of intracellular ROS was detected using the Reactive Oxygen Species Assay Kit (Beyotime Biotechnology, Shanghai, China), and visualized at $E_x/E_m = 488/559$ nm. The area ratios of fluorescence intensity were calculated using the software Image J.

Surface Visualizations: The morphology of sessile cells in the biofilms with and without Al_x HEAs treatment was visualized under SEM (JSM-7001F, JEOL, Japan) and TEM (JEM-2100F, JEOL, Japan). The coupons covered with biofilms were first submerged in a 2.5% (w/w) glutaraldehyde solution for 4 h to fix the sessile cells. Then, each biofilm was dehydrated in 50, 60, 70, 80, 90, 95, and 100% (v/v) ethanol for 8 min at each concentration, respectively. The coupons were finally sputtered with Au before SEM observation to improve the surface conductivity. For SEM and TEM observation, biofilm cells were collected by vibrating vortex for 60 s.^[52] CLSM was also used to observe corrosion morphology and measure pit depth. Then, the morphology and depth of the corrosion pits were analyzed.

Electrochemical Measurements: To assess the influence of Al addition on the corrosion resistance of Al_x HEAs with transient information, a three-electrode system hooked up with an electrochemical workstation (Reference 600 Plus, Gamry, USA) was used. The specimens were cut and mounted in epoxy with copper lines to serve as working electrodes (1 cm^2). Platinum foil was the counter electrode and saturated calomel electrode (SCE) was the reference electrode. At a scan rate of $0.125\text{ mV}\cdot\text{s}^{-1}$, the linear polarization resistance (LPR) was measured from -5 to 5 mV versus the stabilized open circuit potential (OCP). The EIS data were obtained in the frequency range from 10^5 to 10^{-2} Hz with a 5 mV versus E_{OCP} sinusoidal alternate current signal. The potentiodynamic polarization curves were obtained with a scan rate of $0.333\text{ mV}\cdot\text{s}^{-1}$, scanning from -0.25 to 0.5 V (vs OCP) at room temperature only once after incubating for 7 days.

Passive Film Analysis: XPS (ESCALAB250, Thermo VG, USA) was used to investigate the chemical valences and compositions of the passive films formed on the Al_x HEAs after 7 days of immersion. The coupons were placed under a high vacuum and etched with Ar ion bombardment with energy (1486.6 eV , 150 W) to analyze the depth profile. All the results were observed from the sputtering 15 s to eliminate the interference caused by surface impurities. The peaks obtained by XPS were calibrated using the C 1s peak (284.6 eV) and fit with the Avantage software (Version 5.9922).

The TOF-SIMS was acquired using a TOF-SIMS 5–100 instrument (IONTOF GmbH, Germany). to investigate the distribution of the alloying elements present in Al_x HEA samples. A 30 keV Bi^{3+} ion source was used for high-mass resolution spectroscopy. A low-energy electron beam was used for charge compensation. Sputtering was done utilizing 0.5 keV Ar ion at a sputtering rate of $0.3\text{ nm}\cdot\text{s}^{-1}$ (relative to SiO_2). With spectrum acquisition and etching being conducted alternately, consequently, SIMS images were generated and the negative secondary ion counts (intensity) were registered in $0 - 1000\text{ a.m.u.}$ range for all sets of experiments. All the TOS-SIMS results were observed from the sputtering 40 s to eliminate the interference caused by surface impurities. The released Cu ions from Al_x HEAs were recorded using an ICP-OES (ICAP PRO, Thermo Fisher Scientific, USA).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

L.Y. and E.Z. contributed equally to this work. D.X. performed in conceptualization, supervision, and funding acquisition. E.Z. performed in conceptualization, supervision, and writing-review. L.Y. performed in conceptualization, investigation, validation, and writing-original draft. Y.D. performed in validation and writing-original draft. J.H., Z.T., B.L., and Z.Z. performed in validation. Y.W. and Z.L. performed in investigation. M.Z. performed in supervision and writing-review. Y.L. performed in conceptualization and validation; X.L., M.C., B.Y., Y.F., F.W., and T.G. performed in writing-review.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

anti-biofilm, biofilm, high-entropy-alloy, microbiologically influenced corrosion, passive film

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